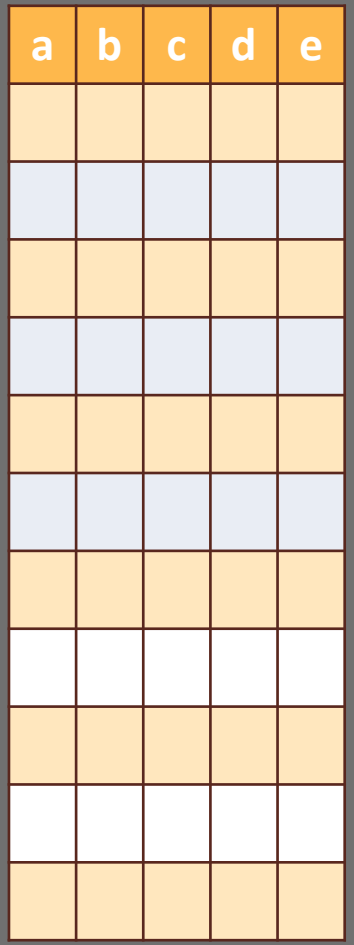
How Do We Find the Desired Data?

How do we find matching tuples?

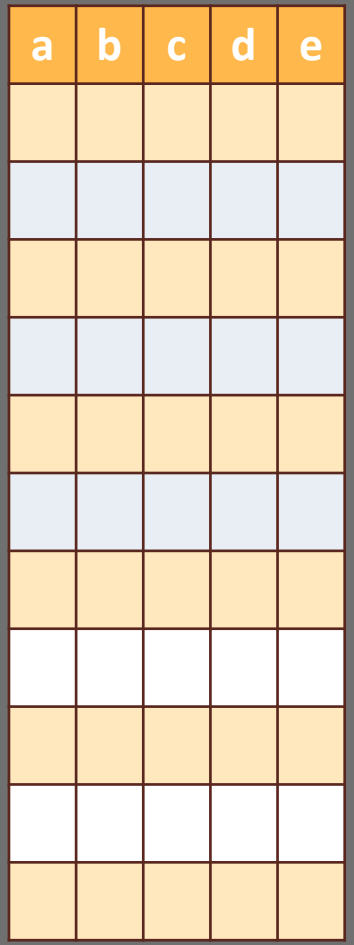
```
SELECT Attributes
FROM   Table
WHERE  Condition
```



Linear Search

```
SELECT …
FROM   Table
WHERE  (b=5 AND d>6) OR e=2
```

$O(\#tuples)$



Binary Search

```
SELECT …
FROM   Table
WHERE  (b=5 AND d>6) OR e=2
```

$O(\log(\#tuples))$

Requires table physically sorted on relevant attributes.

Can only be physically sorted in one order.



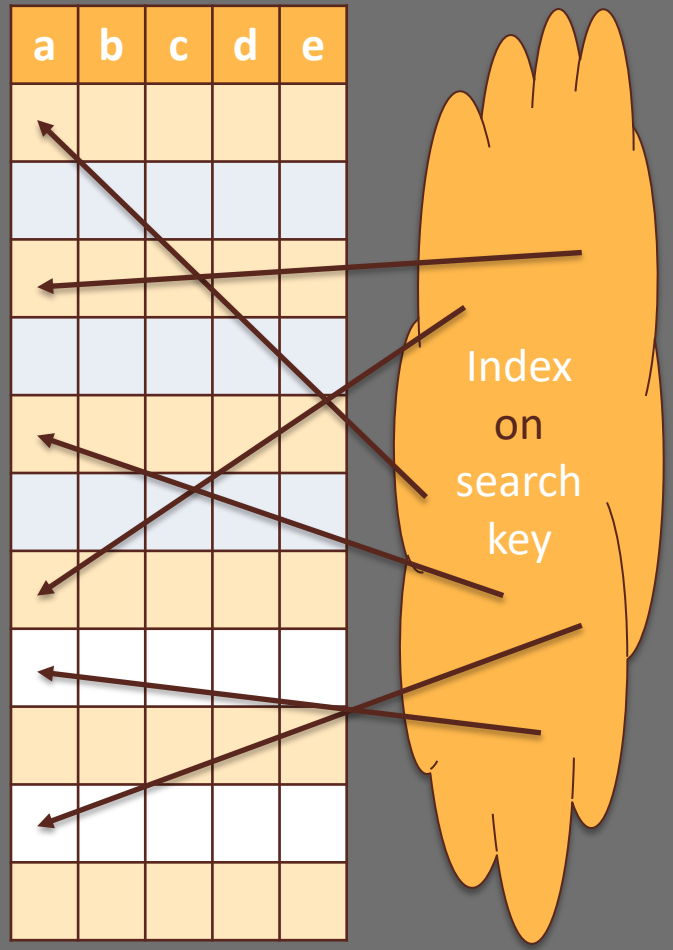
Table clustering

Many:

- Tables clustered on primary key.
- Can specify clustering.

PostgreSQL:

- Tables not maintained in any sorted order.
- Can cluster when desired.



Indexing

```
SELECT …
FROM   Table
WHERE  (b=5 AND d>6) OR e=2
```

$O(1)$ … $O(\log(\#tuples))$

Adds time & space overhead!

Index on what search keys?

Product(id, name, price, category, manufacturer)

If you don’t find it in the index, look very carefully through the entire catalogue.

– Sears, Roebuck, and Co. Consumer’s Guide, 1897